

- ① If $\sin y = x \cdot \cos(x+y)$, then $\frac{dx}{dy}$ is?
- ② The function $f(x) = \begin{cases} \frac{e^{3x} - e^{-5x}}{x}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ is continuous at $x=0$ for what value of k .
- ③ If C_{ij} denotes the cofactor of element P_{ij} of the matrix $P = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & 2 & 4 \end{bmatrix}$, then $C_{31} \cdot C_{23} = ?$
- ④ The function $f(x) = x^2 e^{-x}$ is decreasing in
a) $(0, 2)$ b) $(2, \infty)$ c) $(-\infty, 0)$ d) $(\infty, 0) \cup (2, \infty)$
- ⑤ The principal value of $\tan^{-1}(\tan \frac{9\pi}{8})$ is
- ⑥ In a sphere of radius r , a right circular cone of height h having maximum curved surface area is inscribed. The expression for the square of curved surface area of the cone is?
- ⑦ If the curves $xy = c$ and $y^2 = 4x$ cut at right angles, then the value of c is?
- ⑧ $\int_1^2 \left[\frac{1}{x} - \frac{1}{2x^2} \right] e^{2x} dx$
- ⑨ Solve for x : $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} \quad (x \neq 0)$
- ⑩ Solve the differential equation:
 $x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} + x - y \sin\left(\frac{y}{x}\right) = 0$
Gives that $x=1$ when $y = \pi/2$.